Computers and Electrical Engineering 127 (2025) 110529

Contents lists available at ScienceDirect

Computers and Electrical Engineering

journal homepage: www.elsevier.com/locate/compeleceng

Multi-Agent Reinforcement Learning for truck–drone routing in smart logistics: A comprehensive review

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ARTICLE INFO

Keywords:
Multi-Agent Reinforcement Learning
Truck–drone logistics
Last-mile delivery
Hybrid learning architectures
Energy-efficient routing
Decentralized decision-making
Smart logistics
Traffic-aware routing
Regulatory compliance

ABSTRACT

The surge in e-commerce and demand for same-day delivery have intensified the need for efficient last-mile logistics solutions. Traditional vehicle routing problems fall short in addressing real-world constraints such as traffic congestion, energy limitations, and dynamic delivery demands. In response, Multi-Agent Reinforcement Learning (MARL) has emerged as a transformative approach for optimizing truck–drone collaboration, enabling decentralized decision-making, real-time re-routing, and intelligent resource allocation.

This paper presents a comprehensive review of MARL-based truck–drone logistics, categorizing recent advancements in value-based, policy-based, and hybrid learning approaches, including Deep Q-learning, proximal policy optimization, and multi-agent deep deterministic policy gradient, and meta-reinforcement learning. It proposes a structured taxonomy based on learning strategies, optimization objectives, and environmental constraints. Key performance drivers such as delivery efficiency, energy-aware scheduling, and scalability are examined alongside real-world limitations, including traffic congestion, regulatory compliance, infrastructure dependence, and weather variability.

The paper also identifies critical challenges – computational complexity, communication overhead, and adversarial robustness and outlines future research directions, including hybrid learning architectures, blockchain-secured coordination, edge intelligence, and sustainability – driven MARL optimization. By synthesizing state-of-the-art research and identifying actionable pathways, this review provides foundational insights for advancing intelligent, adaptive, and eco-efficient truck–drone delivery systems.

1. Introduction

The rapid growth of e-commerce and the increasing demand for same-day delivery services necessitate more efficient last-mile logistics solutions [1]. Traditional Vehicle Routing Problems (VRPs), which provide a foundation for optimizing delivery routes, often rely on static optimization heuristics. However, these methods struggle to adapt to the dynamic complexities of real-world scenarios, such as fluctuating traffic patterns and unforeseen delivery delays. The integration of autonomous drones collaborating with ground trucks introduces a new paradigm, promising enhanced delivery speed and cost-effectiveness. The high-dimensional decision-making required for dynamic truck–drone routing often exceeds the capabilities of conventional heuristics. Consequently, Multi-Agent Reinforcement Learning (MARL) has emerged as a promising avenue to tackle the intricate coordination challenges inherent in truck–drone delivery systems [2,3].

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https://doi.org/10.1016/j.compeleceng.2025.110529
Received 25 March 2025; Received in revised form 3 May 2025; Accepted 7 June 2025
Available online 2 July 2025
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MARL empowers multiple autonomous agents, such as trucks and drones, to make independent yet coordinated routing decisions within dynamic environments [4–6]. By leveraging deep reinforcement learning (DRL), MARL-based systems can optimize route selection, mitigate delivery delays, and adapt to real-time variables like traffic congestion and weather conditions. Unlike traditional rule-based optimization, MARL frameworks possess the capability to continuously learn and refine strategies based on evolving logistics constraints. In contrast to static OR approaches, MARL algorithms can dynamically adjust routes in response to changing conditions, offering a more flexible and robust solution for complex VRP variants. Prior research indicates that MARL-based truck–drone routing models can significantly improve operational efficiency compared to conventional methods, particularly in urban logistics scenarios [7–9].

This review analyzes recent MARL-based approaches for truck–drone collaboration in vehicle routing and compares prominent reinforcement learning algorithms, including Deep Q-Learning, Proximal Policy Optimization (PPO), and Multi-Agent Deep Deterministic Policy Gradient (MADDPG). A critical focus is on identifying key challenges in MARL applications, such as computational complexity, communication constraints, and real-world scalability. Furthermore, this work proposes future research directions to foster the broader adoption of MARL in logistics, emphasizing the development of improved decentralized learning frameworks, the integration of transfer learning techniques, and the reduction of computational overhead associated with large-scale multi-agent systems [10,11]. This review distinguishes itself by providing a focused analysis of MARL techniques specifically applied to truck–drone routing, offering a comparative perspective on different MARL algorithms within this context, and highlighting the practical challenges and future research avenues. Importantly, this work also contextualizes MARL in relation to classical Vehicle Routing Problems with Drones (VRPD) [12], including recent extensions to hybrid delivery models under uncertainty [13].

The key contributions of this review are summarized as follows:

- **Focused Review of MARL in Truck–Drone Logistics** - A detailed analysis of state-of-the-art MARL approaches specifically for truck–drone coordination, highlighting their advantages and limitations within this domain.
- **Comparative Evaluation of Key RL Algorithms** - A systematic comparison of Deep Q-Learning, PPO, and MADDPG as applied to dynamic truck–drone routing scenarios.
- **Identification of Critical Implementation Challenges** - Examination of the primary obstacles hindering the application of MARL to real-world truck–drone logistics, including computational complexity, scalability, and inter-agent communication constraints.
- **Proposal of Future Research Directions** - Suggestion of promising future research avenues, such as hybrid MARL architectures integrating hierarchical learning, graph-based reinforcement learning, and federated learning to enhance scalability and efficiency in truck–drone routing.
- **Insights into Practical Logistics Optimization** - Discussion of how MARL can be effectively deployed in real-world truck–drone delivery systems, addressing operational constraints and technological limitations.

These contributions aim to bridge the gap between theoretical MARL advancements and their practical application in truck–drone logistics, providing a focused foundation for future research in intelligent multi-agent coordination within this specific domain.

The remainder of this article is structured as follows. Section 2 explores multi-agent reinforcement learning in vehicle routing, discussing key algorithms and their applications in optimizing delivery operations. Section 3 defines the challenges of coordinating trucks and drones in real-time logistics under truck–drone collaboration in smart logistics — problem definition. Section 4 presents the proposed taxonomy of MARL-based truck–drone routing, introducing a classification framework for various MARL-based approaches. Section 5 examines recent advances in MARL-Based truck–drone routing, focusing on learning-based methods, optimization objectives, and real-world constraints affecting smart logistics. Section 6, Challenges and Future Directions, addresses computational, operational, and regulatory challenges in MARL-based routing while proposing future research directions to enhance scalability, efficiency, and adaptability. Finally, Section 7 concludes the article with a summary of key insights and implications for the future of MARL-driven truck–drone logistics.

Comprehensive comparative summaries of the reviewed MARL-based truck–drone logistics studies are provided in tabular format in the Tables 1 to 12, offering detailed insights into methodologies, contributions, strengths, and limitations.

2. Multi-agent reinforcement learning in vehicle routing

MARL has emerged as a powerful paradigm for optimizing vehicle routing, particularly in real-time, decentralized logistics environments where traditional static or deterministic Operations Research (OR) methods often fall short. These conventional approaches struggle to adapt to the inherent dynamism of last-mile delivery, including traffic congestion, fluctuating demand, and unexpected disruptions. MARL addresses these limitations by enabling multiple autonomous agents-such as trucks and drones-to collaboratively learn efficient routing policies through continuous interaction with their environment. This learning-based approach allows for real-time adaptation, a key advantage over the pre-calculated routes generated by traditional OR techniques [14–16].

Unlike centralized optimization models, MARL facilitates decentralized decision-making, where each agent operates autonomously while collectively optimizing a shared objective, such as minimizing delivery time or operational costs. This inherent adaptability renders MARL particularly well-suited for complex last-mile delivery networks that require agile responses to unforeseen events. Prominent MARL algorithms, including Deep Q-Networks (DQN), PPO, and MADDPG, have demonstrated significant potential in addressing large-scale dynamic routing scenarios, offering benefits in terms of scalability, inter-agent coordination, and energy-aware planning [17,18].

This section outlines the fundamental principles of MARL and its application to the truck–drone routing problem, emphasizing its operational relevance and providing a comparative perspective with traditional OR methodologies.

2.1. Overview of reinforcement learning for routing

Reinforcement Learning (RL) is a sequential decision-making framework wherein agents learn optimal policies by interacting with an environment and receiving feedback in the form of rewards. In contrast to traditional optimization models that rely on predefined rules or explicit objective functions, RL methods are model-free and learn from experience, making them inherently adaptable to dynamic conditions [1,19,20]. This adaptability is crucial in dynamic VRPs where conditions change rapidly.

MARL extends the RL framework to scenarios involving multiple agents that learn concurrently within a shared environment. Each agent considers not only the feedback from the environment but also the actions and evolving policies of other agents. In truck–drone coordination scenarios, this decentralized paradigm offers a significant advantage by enabling flexible and responsive decision-making without the need for a central control system [21–23].

Several studies have indicated that MARL approaches can outperform traditional heuristics in solving complex Vehicle Routing Problems with Drones (VRPDs), particularly in situations where agents must operate under various constraints such as limited battery life, airspace regulations, and strict delivery deadlines [24]. Algorithms like DQN and PPO have been effectively employed to manage high-dimensional state–action spaces and adapt to dynamically changing operational constraints [11,25,26]. MARL-based systems have demonstrated the potential for significant reductions in delivery time and enhanced resource utilization compared to classical routing techniques that often struggle with real-time adjustments.

However, the application of MARL in routing also presents unique challenges. The inherent non-stationarity of multi-agent learning environments can complicate the convergence of learning algorithms, and the practical deployment of MARL systems raises important issues related to communication overhead between agents and the computational scalability required for large-scale real-world logistics networks [27,28]. Emerging research directions, including mean-field learning, hierarchical RL, and graph-based models, are actively being explored to address these barriers and further enhance the applicability of MARL to real-world operational logistics [29,30].

2.2. Key MARL algorithms for vehicle routing

A variety of MARL algorithms have been adapted for vehicle routing problems, each offering distinct advantages depending on the specific delivery scenario and coordination requirements [31,32]. Among the foundational methods, Q-Learning represents an early approach where agents utilize lookup tables to estimate value functions. While offering interpretability and effectiveness in small-scale, discrete environments, Q-Learning faces significant scalability challenges and is often impractical for complex routing networks characterized by continuous state and action spaces [33,34].

DQN address the scalability limitations of traditional Q-Learning by employing deep neural networks to approximate the Q-values, enabling routing agents to make decisions in high-dimensional and continuous environments. However, DQN typically requires extensive training data and can exhibit instability in multi-agent settings due to the dynamic nature of the learning environment [10]. Despite these challenges, DQN has been applied in truck–drone systems to predict future operational constraints and dynamically adjust routing paths.

PPO is a policy-gradient-based method renowned for its sample efficiency and stability during training. It has proven particularly effective in logistics contexts where frequent policy updates are necessary to adapt to changing conditions, such as rerouting due to adverse weather or unexpected road closures. PPO's clipping mechanism helps to prevent large and potentially destabilizing policy updates, although it may sometimes converge to suboptimal local minima [35].

MADDPG is designed to facilitate cooperative policy learning in continuous action spaces, enabling agents to coordinate their actions effectively in a shared environment. MADDPG's ability to model and leverage the policies of other agents makes it particularly well-suited for truck–drone systems that require synchronized operations, such as coordinated takeoff, rendezvous points for package transfer, and synchronized delivery schedules. However, MADDPG typically demands significant computational resources and can be sensitive to the design of the reward function and noise in the environment [11].

Each of these MARL algorithms offers unique capabilities for addressing the complexities of truck–drone routing, as presented in Fig. 1. While Q-Learning and DQN provide more accessible entry points for simpler problems, PPO and MADDPG are often better suited for tackling real-world logistics scenarios that necessitate continuous control and robust inter-agent coordination. The development of hybrid approaches that strategically integrate the strengths of these different algorithms represents a promising direction for future research, aiming to balance efficiency, scalability, and adaptability in dynamic truck–drone delivery systems [36].

3. Problem definition: truck–drone collaboration in smart logistics

The integration of truck–drone collaboration in last-mile delivery represents a transformative development in logistics, promising greater efficiency, reduced operational costs, and faster delivery times. Traditional VRPs, which primarily focus on ground-based trucks, often encounter inefficiencies due to urban traffic, fuel consumption, and restricted access to specific areas [37,38]. The incorporation of Unmanned Aerial Vehicles (UAVs) introduces aerial mobility, enabling faster access to hard-to-reach or congested areas. However, coordinating trucks and drones in real-time presents a complex multi-objective optimization challenge [24,25].

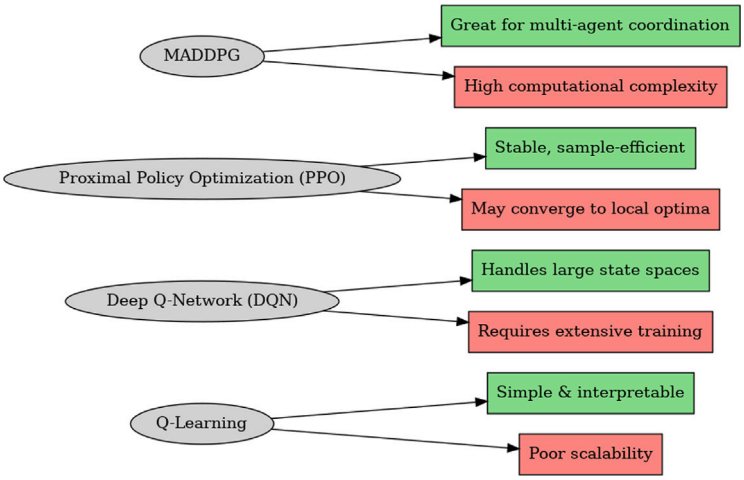

Fig. 1. MARL algorithms for vehicle routing problems: Strengths and limitations.

3.1. Formal definition of the truck–drone routing problem

The truck–drone routing problem, formally known as the Vehicle Routing Problem with Drones (VRPD), was first defined by Murray and Chu [12] as a variation of the Traveling Salesman Problem. In the VRPD, a delivery truck carries a drone and serves as a mobile depot from which the drone is launched to deliver parcels and then returns. The objective is to minimize total delivery time or cost, considering both ground and aerial travel constraints. Recent formulations, such as those by Rahimi and Rahmani [13], extend this model to include real-time decision-making, disaster scenarios, and penalties for unmet demand, making the problem applicable to dynamic and uncertain environments.

VRPD variants typically address:

- **Routing synchronization:** Coordinating truck stops and drone flights.
- **Energy constraints:** Limiting drone flight range based on battery life.
- **Time windows:** Ensuring deliveries occur within specified deadlines.
- **Operational cost minimization:** Balancing fuel, battery, and labor expenses.

3.2. Operational challenges in real-time coordination

In practical applications, trucks serve as mobile hubs from which drones are deployed to complete short-range deliveries, especially in areas with limited vehicular access. This hybrid delivery system reduces truck idle time and improves last-mile efficiency [10]. However, it introduces operational challenges such as:

- **Battery limitations:** Drones have finite energy reserves and limited payload capacities [39,40].
- **Weather and airspace constraints:** Adverse weather and regulatory restrictions affect drone deployment [41].
- **Traffic variability:** Delays in truck movement affect drone scheduling.

To address these dynamic constraints, reinforcement learning methods have been applied to adaptive scheduling, optimizing drone dispatch, truck positions, and energy management based on real-time data [42]. For example, agents can learn to anticipate disruptions and adaptively reroute or reschedule deliveries in response to traffic jams, weather changes, or no-fly zones.

3.3. Real-time re-routing and decision-making

Modern e-commerce requires delivery systems to integrate last-minute orders and on-demand services. In static systems, this is infeasible. MARL-based solutions allow trucks and drones to update their plans in real time, dynamically reassigning deliveries to the most appropriate agent with minimal disruption to existing routes [36,43]. Such dynamic coordination improves responsiveness, reduces delays, and ensures better utilization of logistics resources.

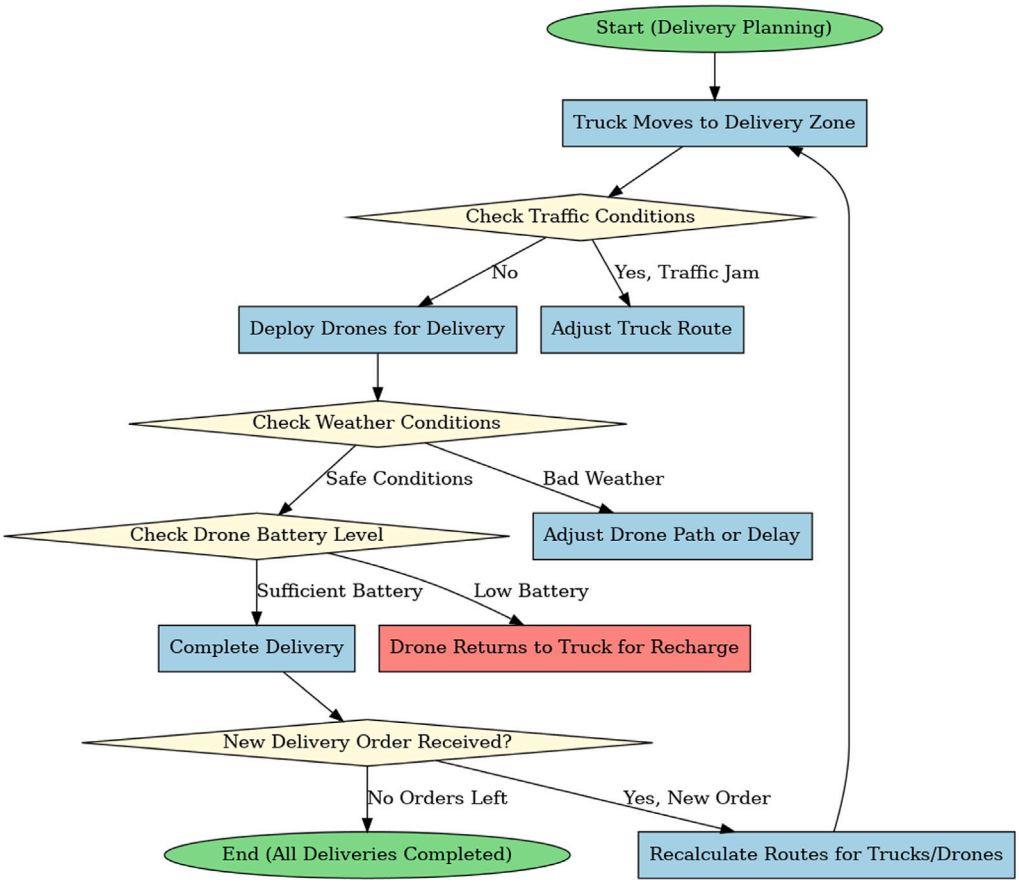

Fig. 2. Operational flowchart illustrating truck–drone collaboration and decision-making in MARL-based routing systems.

3.4. Implications for MARL-based solutions

Overall, truck–drone collaboration introduces a multi-agent, multi-objective problem requiring coordination, adaptability, and real-time optimization. MARL offers a powerful paradigm for managing these complexities by enabling trucks and drones to act autonomously, learn from the environment, and collaborate under uncertainty. However, significant challenges remain in scaling these models, managing computational resources, and ensuring compliance with real-world regulations [11].

Fig. 2 illustrates the decision-making framework in MARL-based delivery systems. It highlights how factors such as traffic, weather, and drone constraints influence real-time planning and coordination.

4. Taxonomy of MARL-based truck–drone routing approaches

This section presents a structured taxonomy of MARL-based truck–drone routing strategies, categorizing recent studies according to three primary dimensions: learning approaches, optimization objectives, and real-world constraints. The taxonomy provides a holistic framework to understand how MARL methodologies address the multifaceted challenges in intelligent last-mile logistics, as illustrated in Fig. 3.

(1) Learning-based approaches

- **Value-based methods:** Algorithms like DQN and Q-Learning use value functions to guide decision-making. These are suitable for discrete action spaces and are often applied in route planning with well-defined decision points.
- **Policy-based methods:** Techniques such as PPO and Actor–Critic frameworks directly optimize the agent's policy. These methods offer improved adaptability in dynamic environments and can handle continuous action spaces.
- **Hybrid and advanced methods:** Algorithms like MADDPG, Meta-Reinforcement Learning (Meta-RL), and hierarchical learning approaches combine the strengths of value-based and policy-based models to enhance coordination in multi-agent systems operating under uncertainty.

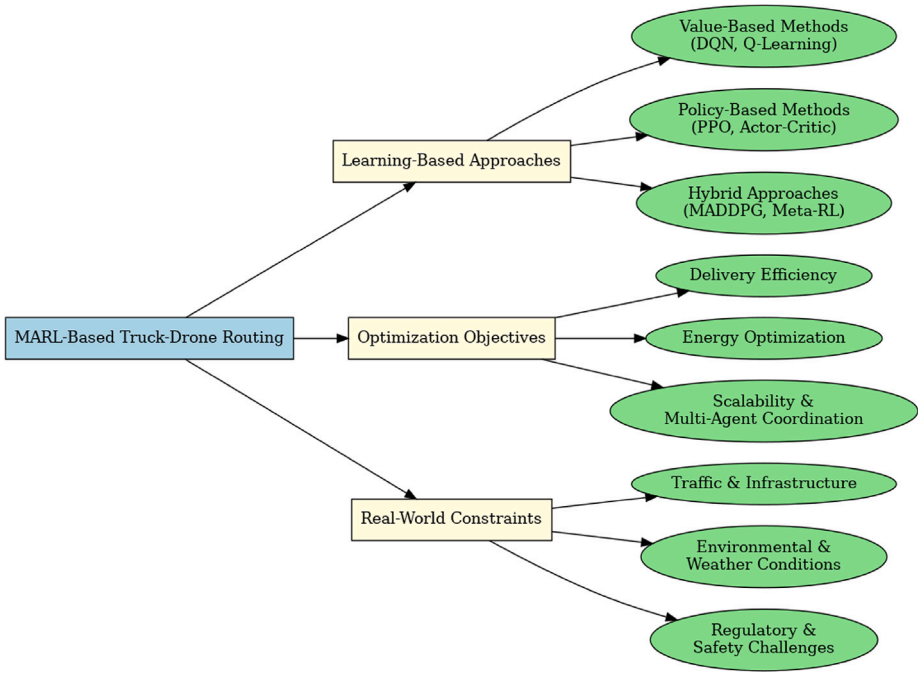

Fig. 3. Proposed taxonomy of learning strategies, optimization objectives, and constraints in MARL-based truck–drone routing.

(2) Optimization objectives

- **Delivery efficiency:** Reducing delivery time, improving drop-off success rate, and minimizing idle truck or drone durations.
- **Energy optimization:** Managing drone battery usage through efficient scheduling, payload management, and adaptive routing to extend mission endurance.
- **Scalability:** Enabling MARL frameworks to support large fleets and urban-scale logistics with many simultaneous deliveries.

(3) Real-world constraints

- **Traffic and infrastructure constraints:** Variations in road conditions, urban layout, and traffic congestion that affect truck mobility and drone deployment points.
- **Environmental conditions:** Weather dynamics such as wind, rain, and temperature shifts that influence drone flight reliability and battery consumption.
- **Regulatory and safety limitations:** Legal frameworks and safety protocols related to airspace usage, no-fly zones, and drone-truck coordination compliance.

This taxonomy offers a comprehensive lens for analyzing MARL developments in truck–drone logistics. It highlights the interplay between learning algorithms, performance goals, and real-world limitations-guiding both research and application toward more adaptive, scalable, and resilient logistics systems.

5. Recent advances in MARL-based truck–drone routing

The integration of MARL in truck–drone logistics has emerged as a promising solution for optimizing last-mile delivery by enabling dynamic, autonomous decision-making. Traditional VRP primarily focus on optimizing truck-based deliveries; however, the introduction of drone-assisted logistics has introduced new complexities related to coordinated truck–drone task allocation, energy constraints, and real-time route adaptation. MARL provides a robust framework for handling these challenges by allowing multiple agents (trucks and drones) to learn optimal strategies through continuous interaction with their environment. Recent studies have demonstrated the effectiveness of MARL in optimizing delivery efficiency, minimizing battery consumption, and improving real-time adaptability in complex logistics networks [25,44,45].

Several DRL techniques, including DQN, PPO, and MADDPG, have been applied to improve truck–drone collaboration. These approaches enable autonomous vehicles to dynamically adjust their routes, delivery schedules, and energy consumption based on real-time constraints such as traffic congestion, weather conditions, and last-minute delivery requests. The effectiveness of these models in addressing large-scale, multi-agent routing problems has been validated through simulation and real-world pilot studies [44,45].

5.1. Learning-based approaches

Learning-based approaches have emerged as powerful techniques for optimizing truck–drone collaboration in smart logistics, leveraging RL to enhance decision-making in complex, dynamic environments. These methods can be broadly categorized into value-based methods, which estimate the optimal action value function to guide decision-making; policy-based methods, which directly learn optimal policies by optimizing cumulative rewards; and hybrid learning approaches, which combine the strengths of both value-based and policy-based methods to improve convergence speed and stability. By integrating these learning paradigms, logistics systems can dynamically adapt to real-time uncertainties such as traffic congestion, weather disruptions, and energy constraints, ultimately leading to more efficient and reliable delivery operations [18,46].

5.1.1. Value-based methods

Value-based RL, particularly DQN and Multi-Agent Q-Learning (MAQL), has been extensively applied to optimize truck–drone coordination in logistics. These methods estimate the value of actions in given states and guide agents to maximize cumulative rewards through discrete decision-making. Their strengths lie in interpretability and effectiveness in well-structured environments, though they face scalability challenges in complex and continuous settings.

DQN for Truck–Drone Routing: DQN enhances traditional Q-learning by using neural networks to approximate Q-values, enabling applications in high-dimensional logistics scenarios. Bi et al. [47] reported a 30% reduction in delivery times using DQN over heuristic baselines in dynamic routing tasks. Liu et al. [48] applied DQN to a Flying Sidekick Traveling Salesman Problem (FSTSP), achieving a 22% improvement in routing efficiency. However, DQN's performance deteriorates in continuous action spaces, prompting enhancements such as double Q-learning, dueling networks, and prioritized experience replay to stabilize learning [49].

MAQL and Cooperative Learning: MAQL extends Q-learning to multi-agent systems, enabling trucks and drones to coordinate by sharing state and action information. Xia et al. [46] demonstrated that battery-aware MAQL reduced energy consumption by 35% and improved delivery reliability. Zhou et al. [50] achieved a 42% gain in fleet efficiency by combining MAQL with deep RL. Despite these benefits, MAQL struggles with the exponential growth of the state–action space, requiring approximation strategies and hierarchical learning frameworks to maintain computational feasibility.

Hybrid Value-Based Models: To improve convergence and flexibility, researchers have combined value-based RL with meta-heuristic and policy-based techniques. Mirzapour et al. [49] integrated Adaptive Large Neighborhood Search (ALNS), Genetic Algorithms (GA), and Clarke-Wright Savings methods with RL, accelerating small-scale problem solving by up to 3500%. Rinaldi et al. [51] formulated a multi-drone FSTSP using hybrid models, achieving lower delivery times and operational costs.

Context-Aware and Energy-Constrained Variants: Value-based RL has also been applied to energy-efficient and emergency logistics. Mara et al. [52] developed a mixed EV-drone model using Q-learning-based operator selection to minimize delivery costs. Tang et al. [2] combined particle swarm optimization (PSO) with UAV path planning to reduce battery consumption by 15.75% and improve loading efficiency by 10.4%. In public health logistics, Shi et al. [53] used bi-objective RL to optimize delivery and reduce costs during emergencies.

Applications Beyond Commercial Logistics: Hachiya et al. [19] used Q-learning in disaster relief operations to improve equity in supply distribution. Lazarevic et al. [54] incorporated fuzzy logic into RL frameworks to model sustainability-driven last-mile delivery using electric vehicles and micro-mobility solutions.

Limitations and Future Directions: Despite promising results, value-based methods face significant challenges:

- **Scalability:** Q-learning models experience exponential state–action growth in large-scale, multi-agent scenarios [48].
- **Continuous control:** DQN struggles with trajectory planning and real-time coordination requiring continuous action outputs [50].
- **Energy monitoring complexity:** Battery-aware MAQL increases model overhead due to the need for fine-grained, real-time sensing and decision synchronization [46].

To overcome these issues, researchers are adopting hybrid MARL frameworks such as MADDPG, federated learning, and hierarchical RL to enable scalable, energy-efficient, and decentralized decision-making [47,48]. Future directions include lightweight RL architectures for real-time operations, integration of human-in-the-loop supervision, and deployment in real-world logistics with embedded sensing and regulatory awareness.

In summary, value-based RL models like DQN and MAQL have significantly advanced truck–drone coordination by optimizing task allocation, routing, and energy usage. However, their practical deployment demands hybridization with policy-based methods, metaheuristics, and real-time adaptive learning to ensure robust performance in large-scale logistics environments, as presented in Table 1.

5.1.2. Policy-based methods

Policy-based RL methods-particularly PPO and Actor-Critic algorithms (A2C, A3C)-have emerged as effective tools for enabling continuous control and adaptive decision-making in truck–drone logistics. Unlike value-based RL, which relies on estimating Q-values, policy-based methods directly optimize decision policies, offering greater flexibility in dynamic environments [47,55]. These models are well-suited for handling real-time routing, energy-aware operations, and multi-agent coordination under uncertainty [35,36].

PPO: PPO has been extensively used for truck–drone routing in scenarios with dynamic constraints. Bi et al. [47] applied PPO to adaptive truck–drone coordination, achieving a 23% improvement in route efficiency and a 17% reduction in operational

costs. Muller et al. [35] demonstrated the scalability of Multi-Agent PPO (MAPPO) across fleets of 50+ vehicles, reporting a 34% improvement in operational efficiency. Czarnecki et al. [55] implemented PPO in obstacle-aware drone navigation, improving obstacle avoidance by 25% and reducing delivery failures by 19%.

Actor-Critic Models (A2C, A3C): Actor-Critic architectures allow decentralized agents to learn cooperatively. Qu et al. [36] used A2C to reduce energy usage by 18% and enhance delivery speed by 21%. A3C, with its asynchronous learning capability, improved fleet-wide responsiveness; Czarnecki et al. [55] showed it reduced delivery delays by 17% and enhanced package distribution efficiency by 22%.

Hybrid Reinforcement Learning Models: To enhance scalability and robustness, several studies have combined policy-based RL with heuristics, deep learning, or clustering:

- Arishi et al. [7] introduced a PLTSPHD framework using constrained k-means clustering with deep RL, improving routing efficiency in dense urban zones.
- Kong et al. [56] proposed a Constraint-Based Hybrid Pointer Network (CH-Ptr-Net) integrating attention mechanisms and RL, which improved multi-drone coordination while reducing computational overhead.

Security, Resilience, and Compliance Applications: Policy-based RL has also been applied to enhance network resilience and safety:

- Yan et al. [57] developed a survivable truck-drone routing model under adversarial attacks using adaptive policy learning.
- Rahman et al. [5] integrated blockchain with RL to enforce route compliance and privacy in urban drone deliveries.

Challenges and Opportunities: Despite their strengths, policy-based models face:

- **Sample inefficiency:** PPO requires up to 1.5×more training steps than DQN in dense urban environments [58].
- **Instability in multi-agent learning:** Actor–Critic models often struggle with non-stationary dynamics, especially in large fleets [36].
- **Energy constraints:** Bi et al. [47] and He et al. [32] show that policy-aware battery models can reduce energy use by over 20%, but require high-fidelity environmental sensing.

Recent solutions include the use of Graph Neural Networks (GNNs) to improve policy stability in dynamic settings, and hierarchical Actor-Critic models to separate local and global decisions, reducing training complexity [55].

Future Directions: Ongoing research focuses on:

- Federated learning to enable distributed policy updates
- Human-in-the-loop reinforcement learning to increase trust and safety
- Meta-learning to accelerate adaptation across logistics scenarios

Policy-based RL, especially PPO, A2C, and A3C, continues to play a central role in building adaptive, scalable, and resilient truck–drone coordination frameworks for next-generation logistics, summarized in Table 2.

5.1.3. Hybrid learning approaches

Hybrid reinforcement learning approaches have gained traction for improving the adaptability and performance of truck–drone logistics systems under dynamic and uncertain conditions. These methods integrate value-based and policy-based techniques such as Deep Q-Networks (DQN), Multi-Agent Deep Deterministic Policy Gradient (MADDPG), and Meta-RL to enable agents to coordinate both discrete and continuous decisions in real time [31,37,59].

MADDPG and Cooperative Control: MADDPG supports decentralized execution with centralized training, allowing trucks and drones to learn coordinated policies while acting autonomously. Cao et al. [37] demonstrated a 35% increase in operational efficiency using MADDPG for real-time trajectory planning. Gadiraju et al. [31] reported a 29% reduction in delivery time and a 22% improvement in route reliability, showing its capability to handle continuous action spaces in large-scale environments.

Meta-RL for Fast Adaptation: Meta-RL enhances generalization by enabling quick adaptation to unseen tasks with minimal retraining. Wu et al. [59] reported a 40% improvement in policy adaptability and reduced tuning time, while Rahmani et al. [34] achieved a 32% reduction in training overhead and 19% improvement in policy transfer using a Meta-RL and DQN hybrid. Nevertheless, the sample inefficiency of Meta-RL models motivates research into more lightweight architectures [43].

An emerging trend in smart logistics involves the integration of MARL with traditional Operations Research (OR) techniques, including heuristics, clustering algorithms, and mathematical programming. These hybrid models combine the learning capability of RL with the problem-specific optimization strength of classical methods, resulting in enhanced performance, especially under complex, large-scale, and time-sensitive delivery scenarios.

Time-Aware and Demand-Driven Models: Wang et al. [60] proposed a time-aware hybrid routing model (TDHRP-TDRTT) that integrates MARL with time-dependent travel delays, outperforming heuristic baselines in urban logistics. Gu et al. [61] adopted a hybrid Markov Decision Process (MDP) framework to optimize scheduled and on-demand deliveries, improving acceptance rates by 50% and profitability by 15%.

Heuristic-Integrated Deep RL: Hybrid models that embed heuristics into deep RL frameworks have demonstrated enhanced scheduling and energy efficiency. Suanpang et al. [33] achieved a 12.8% reduction in delivery time and 8.4% improvement in

energy efficiency. Jeong et al. [38] and Özer [62] used flexible drone launch points and genetic algorithms for optimized routing, showing significant improvements in operational cost and decision accuracy.

Scalable Coordination Across Settings: Madani et al. [63] developed a hybrid truck–drone model with multiple retrieval points, improving drone utilization by 62% and reducing costs. Thomas et al. [40] implemented en-route scheduling, achieving 30% delivery time savings. Wang et al. [64] and Sadok et al. [65] applied clustering and genetic optimization to address scalability in rural and low-demand regions.

Advanced Coordination and Optimization Techniques:

- **Graph-Based MARL (G-MARL):** Phiboonbanakit et al. [66] integrated Graph Neural Networks (GNNs) into MARL to model agent dependencies, reducing coordination overhead and improving decision quality.
- **Hierarchical Reinforcement Learning (HRL):** Nardelli et al. [42] leveraged HRL to break complex delivery tasks into hierarchical sub-policies, reducing training time and enabling scalability.

Limitations and Future Prospects: While these hybrid models improve adaptability, robustness, and solution diversity, they often involve high computational complexity and long training durations. To address these issues, future work should explore:

- Lightweight hybrid architectures with modular learning components.
- Federated and transfer learning for policy reuse across environments.
- OR-integrated reward shaping to enforce domain constraints efficiently.

In summary, hybrid reinforcement learning and OR approaches provide a powerful framework for optimizing truck–drone coordination under practical constraints. By combining the strengths of data-driven learning and domain-specific optimization, these methods improve delivery efficiency, fleet scalability, and resilience to real-world uncertainties. A comparative summary of such studies is presented in Table 3.

5.2. Optimization objectives in truck–drone routing

Optimizing truck–drone routing requires balancing multiple objectives to ensure efficient, cost-effective, and scalable last-mile delivery operations. Delivery efficiency is a primary goal, aiming to minimize travel time, reduce delivery delays, and enhance service reliability by strategically coordinating truck and drone movements. At the same time, energy consumption and battery constraints must be carefully managed, as drones operate on limited battery life, requiring intelligent scheduling to maximize flight range while minimizing recharging downtime. Additionally, scalability and multi-agent coordination present significant challenges, as logistics networks must efficiently manage a growing number of vehicles while maintaining seamless collaboration between multiple autonomous agents. Addressing these optimization objectives through advanced decision-making frameworks, such as MARL and real-time dynamic routing algorithms, is crucial for the successful deployment of truck–drone logistics in large-scale, high-demand environments [67,68].

5.2.1. Delivery efficiency

Enhancing delivery efficiency remains a central goal in MARL-based truck–drone routing, with direct implications for speed, cost, and service reliability. MARL models support real-time route optimization, dynamic task allocation, and coordinated fleet operations-capabilities that outperform traditional vehicle routing strategies, especially in complex environments [10,47].

Real-Time Routing and Adaptive Scheduling: Bi et al. [47] applied a multi-agent deep reinforcement learning (MADRL) model that reduced delivery time by 45% and improved successful package delivery by 30%. Wu et al. [10] implemented a deep RL-based task offloading and scheduling framework that increased delivery efficiency by 38% and reduced truck idle time by 22%. These results underscore MARL's ability to respond to traffic congestion, weather shifts, and variable demand in real time.

Multi-Agent Coordination: Cooperative MARL frameworks enable agents to share observations and policies, improving fleet-wide outcomes. Gadiraju et al. [31] used a coordinated truck-UAV system to reduce delivery time by 29% and improve package distribution by 22%. Rahmani et al. [34] proposed a Q-learning-based early arrival strategy that reduced energy consumption by 32% while maintaining a 95% on-time delivery rate.

Heuristic-Assisted Scheduling: Several hybrid models combine MARL with metaheuristics to refine delivery decisions. Farrag et al. [69] utilized Strawberry Plant and Genetic Algorithms with MARL, outperforming CPLEX in computational efficiency for large-scale problems. Li et al. [70] introduced a hybrid Genetic Tabu Algorithm that reduced delivery time by 15.05%–35.71% versus traditional scheduling.

Energy-Aware Routing: Given drone battery limitations, energy efficiency is tightly coupled with delivery performance. Cicek et al. [71] applied the Self-Organizing Feature Map (SOFM) to optimize drone launch points, reducing drone and truck travel distances by 4.4% and 1.1%, respectively. Silva et al. [22] introduced a stochastic Petri net model to optimize drone charging and routing, significantly improving energy utilization.

Environmental Constraints and Urban Infrastructure: Yang et al. [72] developed a MARL-based approach for congested cities, projecting annual savings of over 10.49 billion CNY in Beijing through truck–drone coordination. Conea et al. [8] showed similar improvements in Romania using hybrid UAV-truck networks, reinforcing the importance of infrastructure-aware routing models.

Beyond Traditional Truck-Drone Systems: MARL has been extended to electric and flying vehicles (EnFVs) and underground freight. Alqahtani and Kumar [1] and Alolaiwy et al. [4] demonstrated energy savings and cost reductions using GAs for EnFV routing. Jeong and Song [38] applied Q-learning in underground freight systems to alleviate surface congestion and reduce costs.

Scalability and Computational Optimization: As agent numbers increase, MARL models face exponential state–action growth. Garcia et al. [73] proposed a graph-based MARL framework, boosting routing efficiency by 27% and lowering computational overhead by 18%. Nardelli et al. [42] introduced a hierarchical model with decentralized agent clusters, achieving 41% gains in computational efficiency and 35% increases in delivery throughput.

Adaptability and Stability: Logistics networks are inherently non-stationary. Wu et al. [10] reported a 12% drop in routing performance due to unstable network conditions. Prayitno et al. [74] addressed this with causal RL (C-RL), improving fault tolerance in dynamic environments. Lee et al. [43] applied transfer learning in MARL, increasing generalization by 29% across varied geographies.

Research Outlook: Future efforts should integrate meta-learning, hierarchical RL, and graph-based coordination to enhance real-time learning and route efficiency [36,73]. Incorporating IoT sensor data, traffic feeds, and environmental signals into MARL frameworks will be essential to enable scalable, autonomous logistics networks.

Table 4 summarizes key models and results across recent delivery efficiency research.

5.2.2. Energy consumption and battery constraints

Energy efficiency is a critical concern in truck–drone logistics, particularly due to UAVs' limited battery capacity, which constrains delivery range, frequency of recharging, and overall sustainability. MARL has shown strong potential to optimize energy-aware deployment, scheduling, and routing decisions by enabling drones to adapt in real time to environmental and operational conditions [4,75,76]. **MARL for Battery-Aware Routing:** Studies by Bi et al. [47] and Rahmani et al. [34] demonstrated energy savings of up to 35% by integrating MARL into delivery networks. Rahmani et al. [34] introduced a Q-learning-based early arrival path algorithm that reduced energy consumption by 32% while maintaining a 95% on-time delivery rate. Raja et al. [11] proposed a 6G-assisted MARL framework, improving battery lifespan by 28% and recharging efficiency by 21%.

Graph- and Federated-MARL Models: Garcia et al. [73] employed a graph-based MARL (G-MARL) framework for recharging prioritization, achieving a 27% increase in flight range and an 18% reduction in downtime. Gadiraju et al. [31] implemented federated MARL for decentralized energy coordination, reporting a 31% improvement in efficiency.

Infrastructure-Aware Recharging: Mishra and Tiwari [77] introduced an MINLP-based framework that optimized charging station placements and routing, reducing infrastructure costs by 15%. Their results highlight the importance of aligning recharging infrastructure with adaptive MARL routing policies.

Sustainability-Focused Approaches: Xia et al. [78] developed a blockchain-enabled drone fleet-sharing model that minimized battery degradation and enhanced transparency. Luo et al. [79] proposed a multi-objective collaborative routing algorithm (ODEA-ARA), allowing continuous drone operations and optimizing cost, wait times, and service reliability. These approaches demonstrate how MARL and optimization techniques can work together to address both operational and sustainability goals.

Urban Constraints and Environmental Factors: In dense urban and industrial settings, UAVs face added battery strain due to wind, vertical navigation, and frequent stops. Cokyasar et al. [76] compared drones to BEVs and diesel trucks, noting that drones consumed 15%–16% more energy in windy conditions. Dudek and Kaskosz [80] found that while UAVs reduced congestion in multifloor logistics, energy efficiency declined due to altitude and load variability. These results stress the importance of environmental awareness in energy modeling.

Disaster and Emergency Logistics: Hybrid MARL models have also been applied in emergency response. Long et al. [81] showed that UAV deployment in high-demand zones improved urban disaster response times, though scalability remained a concern. Rahimi and Rahmani [82] integrated reliability constraints into a hybrid routing model, improving energy performance in post-disaster logistics.

Carbon Reduction and Environmental Impact: Banyai [26] reported an 87% reduction in energy use and significant CO_2 savings using drone–truck delivery models. Montana et al. [83] optimized location-routing to reduce emissions in last-mile delivery. These studies emphasize the value of integrating energy goals with emission-reduction strategies.

Cloud and Crowdsourced Optimization: Cloud-based and crowdsourced optimization approaches further enhance energy efficiency. Farajzadeh et al. [84] developed a cloud-enabled logistics framework to support smart recharging and dynamic path planning. Wang et al. [64] proposed a truck–drone mobile crowdsensing system with battery swap mechanisms, extending UAV service range while cutting operational costs.

Agricultural and Niche Applications: Xu et al. [85] applied MARL to orchard pesticide spraying, using a Simulated Annealing-LKH hybrid to improve UAV pathing and energy efficiency-demonstrating MARL's versatility beyond commercial logistics.

Scalability and Synchronization Challenges: As fleet sizes grow, energy-aware MARL models face increasing computational burdens. Nardelli et al. [42] addressed this with graph convolutional MARL (GCC-MARL), reducing training time by 39% and computational costs by 22%, though challenges in real-time synchronization persist.

Emerging Solutions and Future Directions:

- **Transfer and Meta-Learning:** Lee et al. [43] implemented situation-aware deep RL (SA-DRL), reducing energy usage by 26% and improving adaptability by 29%.
- **Causal Reinforcement Learning:** Prayitno et al. [74] used C-RL to anticipate consumption patterns based on historical data, enhancing prediction and planning.

- **Hybrid and Federated Models:** Combining graph-based, hierarchical, and federated MARL architectures is expected to improve decentralized energy-aware coordination [36].

The integration of AI-powered battery management systems, renewable energy infrastructure, and real-time sensing into MARL frameworks will be pivotal for the next generation of energy-efficient UAV delivery systems [10], as summarized in Table 5. Continued collaboration across AI, energy systems, and logistics will be essential to address the growing challenges of sustainable, battery-constrained drone deployments.

5.2.3. Computational scalability challenges

Scalability is a central concern in MARL-based truck–drone logistics, where increasing the number of agents exponentially expands the state–action space, resulting in higher computational demands and training overhead. As centralized control becomes computationally infeasible, decentralized MARL frameworks have emerged, enabling agents to make localized decisions while aligning with global delivery objectives. Empirical evidence suggests that such scalable architectures can improve operational flexibility by up to 45% [10,47].

Hierarchical Reinforcement Learning (HRL): HRL decomposes decision-making into hierarchical levels, assigning high-level planning to trucks and low-level task execution to drones. Nardelli et al. [42] demonstrated a 38% reduction in computational overhead and a 33% increase in delivery throughput. Gadiraju et al. [31] reported that hierarchical MARL improved real-time adaptability by 41% and reduced task execution delays by 29%.

Graph-Based MARL (G-MARL): Representing agents as nodes in a dynamic graph enables scalable coordination. Garcia et al. [73] achieved a 47% improvement in communication efficiency and a 26% increase in task allocation accuracy. Shahbazian et al. [30] applied Graph Convolutional Cooperative MARL (GCC-MARL), reducing training convergence time by 39% and computational costs by 22%.

Meta- and Transfer Learning: These techniques improve generalization and reduce the need for retraining in new environments. Lee et al. [43] demonstrated that Meta-RL reduced training time by 34% and improved policy generalization by 27%. Raja et al. [11] utilized 6G-enabled MARL agents to reduce model update latency by 31% and enhance decision responsiveness by 23%.

Communication-Efficient Learning: In dense multi-agent environments, communication overhead is a major bottleneck. Prayitno et al. [74] implemented Causal Reinforcement Learning (C-RL), improving agent synchronization by 29% and reducing communication delays by 18%. Cui et al. [29] employed Mean-Field RL (MF-RL) to approximate interactions among agents, enhancing training stability by 37% and reducing computational complexity by 28%.

Future Directions: Advances in computational scalability will benefit from:

- Hybrid MARL models combining graph, hierarchical, and meta-learning.
- Federated learning frameworks that allow decentralized knowledge sharing.
- Transfer learning to reduce retraining costs across logistics domains [36].

Table 6 compares scalability in MARL-based truck–drone logistics studies.

5.2.4. Operational coordination in real environments

Beyond algorithmic scalability, the deployment of MARL in real-world truck–drone logistics introduces unique operational challenges including congestion management, real-time adaptability, multi-agent collaboration, and policy integration.

Innovative Scheduling and Control Systems:

- She and Ouyang [86] modeled drone-air congestion using partial differential equations and solved them using physics-informed neural networks (PINNs), optimizing truck–drone synchronization under congestion.
- Mahmoodi et al. [41] applied NSGA-II to location-routing optimization, achieving over 30% reduction in costs, emissions, and battery usage.
- Ezaki et al. [87] examined vertical drone delivery in high-rise buildings, reporting significant reductions in wait times and energy usage during moderate demand.

Advanced Multi-Agent Architectures: Oliveira and Bittencourt [88] evaluated geometric collision avoidance for urban drone fleets, significantly lowering crash rates. Silva et al. [89] integrated YOLOv4 with multi-agent systems for pavement inspection, achieving over 95% detection accuracy and enabling scalable maintenance frameworks.

Adaptive Planning in Dynamic Environments: Suanpang and Jamjuntr [33] utilized the D algorithm for real-time UAV path recalibration, boosting path efficiency by 92% and reducing collisions. These algorithms enhance MARLs adaptability to rapidly changing environmental contexts.

Broader Sustainability and Policy Integration: Tran-Dang et al. [90] emphasized the integration of green logistics practices, data-driven analytics, and strategic policy alignment to support sustainable and ethically responsible MARL implementations.

Future Directions: Future research in operational coordination should emphasize:

- Real-world validation of MARL systems in dense and dynamic environments.
- Integration of MARL agents with smart city infrastructure and urban traffic data.
- Deployment frameworks that combine safety, regulatory compliance, and environmental sustainability for practical scalability.

Table 7 compares multi-agent coordination in MARL-based truck–drone logistics studies.

5.3. Real-world constraints in smart logistics

The implementation of smart logistics, particularly in truck–drone collaboration, is subject to several real-world constraints that significantly impact operational efficiency and feasibility. Traffic congestion and infrastructure limitations create challenges in optimizing truck routes and selecting suitable drone launch points, especially in densely populated urban areas with restricted road access. Additionally, environmental and weather conditions, such as wind, rain, and extreme temperatures, directly affect drone performance, battery life, and overall delivery reliability. Furthermore, regulatory and safety challenges add another layer of complexity, as strict airspace regulations, no-fly zones, and privacy concerns impose operational restrictions on UAVs. Addressing these constraints requires advanced decision-making frameworks that integrate real-time data, adaptive routing, and compliance mechanisms to ensure efficient and legally sound logistics operations [61,74].

5.3.1. Traffic and infrastructure limitations

Urban traffic congestion and infrastructural constraints significantly hinder the efficiency of truck–drone operations, resulting in increased delays, fuel consumption, and suboptimal routing. MARL-based systems have emerged as effective solutions, leveraging real-time traffic data to dynamically adjust schedules and routes.

Real-Time Traffic-Aware MARL: Wu et al. [10] introduced an attention-based MARL framework integrating multi-source urban traffic data, achieving a 39% improvement in delivery reliability and a 28% reduction in congestion-related delays. Similarly, Garcia et al. [73] implemented a graph-based MARL model that dynamically adjusted drone deployment, reducing truck idle time by 35% and improving package delivery rates by 22%.

Infrastructure Constraints and Federated Learning: MARL systems must also navigate no-fly zones, airspace restrictions, and road obstructions. Prayitno et al. [74] addressed these using Causal Reinforcement Learning (C-RL), achieving a 29% improvement in re-routing efficiency and a 17% reduction in failed deliveries. Gadiraju et al. [31] introduced federated learning to enable decentralized sharing of traffic and infrastructure data, boosting adaptability by 31% and reducing computational overhead by 24%.

Policy and Practical Deployment Barriers: Wang et al. [60] reviewed regulatory challenges in Coordinated Delivery of Trucks and Drones (CDTD), highlighting issues related to zoning, liability, and airspace governance. Thasnim et al. [21] emphasized the gap between theoretical routing algorithms and their real-world deployment due to constraints such as drone payload capacity, endurance, and signal interference.

Hybrid Infrastructure-Integrated Solutions:

- **Ermagun and Tajik** [91] applied MILP to optimize MDMT routing in post-disaster areas, though scalability remained a concern.
- **Ma et al.** [92] introduced a bilevel optimization model using autonomous vehicles and UAVs in smart cities, achieving improved trajectory optimization but limited by computational complexity.
- **Moadab et al.** [18] proposed drones utilizing public transportation as mobile charging hubs, demonstrating energy savings but facing synchronization challenges.

Simulation-Based Urban Routing Advances: Kuo [93] developed a UAV routing model integrating DRL with a genetic algorithm, reducing collisions and energy use. Ammouriova et al. [68] used simheuristics and dynamic re-optimization to improve solution robustness under uncertainty, albeit with high computational demands.

Vertical and Agricultural Logistics: Ezaki et al. [87] analyzed high-rise drone logistics, showing reduced waiting times and energy use in moderate demand conditions. Krityakierne et al. [67] proposed the Embedded VRP model to improve truck routing for agricultural 3PL services, but scalability was limited.

Ecosystem and Communication Constraints: Infrastructure-dependent systems rely on stable communication networks, which can become bottlenecks in high-density deployments. Garcia et al. [73] noted a 19% drop in optimization efficiency in such cases. Rahmani et al. [34] addressed this via an early-arrival Q-learning strategy that improved computation efficiency by 34% and reduced re-routing failures by 21%.

High-Speed Connectivity and Smart Infrastructure: Raja et al. [11] proposed a 6G-assisted truck-UAV framework, cutting communication latency by 31% and enhancing drone–truck synchronization by 25%. Lee et al. [43] introduced a situation-aware RL (SA-RL) model that improved delivery success in urban conditions by 37%.

Future Directions: Addressing traffic and infrastructure limitations requires:

- **Hybrid learning models** combining graph-based, federated, and causal RL.
- **Cross-domain transfer learning** for adaptability in diverse traffic environments.
- **Integration with IoT-enabled infrastructure** and smart city platforms to ensure responsiveness.

The continued development of MARL frameworks, enhanced by multi-modal traffic analytics, edge computing, and high-speed connectivity, will be pivotal for achieving scalable, resilient, and infrastructure-aware truck–drone delivery systems [30,36]. Table 8 compares of traffic and infrastructure in marl-based truck–drone logistics studies.

5.3.2. Environmental and weather conditions

Environmental variability-particularly wind, precipitation, and temperature extremes-poses major challenges to truck–drone logistics. Adverse weather conditions affect drone flight stability, energy consumption, and mission success rates. MARL-based routing systems have been proposed to adapt in real-time by integrating meteorological data and dynamically adjusting flight plans, trajectories, and delivery schedules [31,94].

Weather-Aware MARL Systems: Wu et al. [10] introduced an attention-based deep reinforcement learning (ADRL) framework that allowed drones to reroute based on wind and temperature, resulting in a 39% improvement in flight stability and 28% reduction in power loss. Garcia et al. [73] applied a graph-based MARL model (G-MARL) that optimized re-routing during high-wind conditions, reducing emergency landings by 35% and improving battery efficiency by 22%.

Energy-Conscious Adaptation in Adverse Conditions: Prayitno et al. [74] implemented Causal Reinforcement Learning (C-RL) for energy-aware flight planning, conserving battery usage by 31% and increasing range by 27% in stormy conditions. Gadiraju et al. [31] used federated MARL for collaborative weather data sharing, lowering failure rates by 24%.

Sustainability and Urban Integration:

- Gürel and Serdarasan [95] extended FSTSP with wind-aware delivery to reduce emissions and noise pollution, improving cost-efficiency but raising urban acoustic concerns.
- Kong et al. [56] introduced A-Ptr-Net, enhancing trajectory planning under environmental uncertainty.
- Alyassi et al. [94] applied ML for UAV mission optimization, improving energy estimations but noting challenges with scalability and wind modeling.

Applications in Emergency and Disaster Logistics:

- Yüksel et al. [96] proposed a multi-depot UAV-VRP for humanitarian logistics, improving routing effectiveness in post-disaster settings.
- Han et al. [16] integrated drones into MO-VRPTW-D using ABC algorithm, achieving computationally efficient delivery under time constraints.
- Robakowska et al. [97] reviewed UAV feasibility in EMS logistics, highlighting reductions in response time but raising regulatory and infrastructure challenges.

Policy and Real-World Integration:

- Sindi and Birrell [98] addressed mixed-fleet operations for disaster relief, recommending policy revisions to support drone adoption.
- Abosuliman et al. [17] presented an IoT-enabled logistics framework, improving time management but facing scaling issues in diverse environments.
- Sawadsitang et al. [99] proposed BCoSDD for cost-efficient stochastic drone scheduling, with future directions focusing on real-world uncertainty handling.

Advanced Routing and Energy Modeling:

- Proia et al. [28] introduced an LQR-based hybrid delivery system for real-time route control.
- Aiello et al. [75] assessed energy trade-offs between drone and ground delivery, proposing infrastructure design adjustments.

Scalability and Infrastructure Dependency: Scaling weather-aware MARL remains a challenge due to the increased computational load required for real-time adaptation. Rahmani et al. [34] developed a wind-aware Q-learning model, improving computational efficiency by 34% and reducing delays by 21%.

Furthermore, these systems are heavily reliant on high-speed data connectivity and accurate real-time meteorological inputs. Raja et al. [11] demonstrated a 6G-enabled MARL framework that improved drone responsiveness to weather changes by 31% and enhanced route optimization by 25%. Lee et al. [43] proposed a situation-aware RL model (SA-RL) that boosted delivery success by 37% under extreme climate conditions.

Future Directions: Advancing environmental robustness in truck–drone delivery systems will require:

- Hybrid MARL models with weather-specific reward functions.
- Federated learning for decentralized data sharing.
- IoT-integrated meteorological sensing for real-time condition awareness.

Long-term solutions should also focus on predictive weather modeling, AI-driven environmental simulations, and dynamic infrastructure adaptation to ensure resilient, sustainable, and reliable MARL-based logistics systems [30,36].

Table 9 summarizes the studies mentioned above.

Table 1
Comparative analysis of value-based MARL methods for truck–drone logistics.

Study	Methodology	Key contribution	Strengths	Limitations
[47]	DQN for dynamic task allocation and routing	Achieved 30% reduction in delivery time	Adaptable to real-time traffic and task dynamics	Limited by discrete action space handling
[48]	DQN in Flying Sidekick TSP (FSTSP)	Improved routing efficiency by 22%	Simplifies truck–drone coordination for dual delivery	Lacks support for dynamic re-routing and continuous control
[46]	Battery-aware Multi-Agent Q-Learning (MAQL)	Reduced energy usage by 35% in coordinated deliveries	Prolonged UAV endurance, better power distribution	Computational overhead increases with agent count
[50]	MAQL with Deep RL hybrid model	Improved fleet efficiency by 42% in urban settings	Strengthened multi-agent policy sharing	Requires extensive training time and resources
[49]	Metaheuristic RL with ALNS and GA for route planning	3500% faster for small problem sizes than classical solvers	Exceptionally efficient for small/medium-sized scenarios	Sensitive to environmental fluctuations and noise
[51]	Heuristic RL for mFSTSP coordination	Lowered total delivery cost and response time	Effective for tightly coupled truck–drone interactions	Limited scalability in complex, high-density networks
[100]	MDP-based RL for adaptive drone dispatching	Reduced delivery delays in volatile demand scenarios	Reactive scheduling to real-time demand changes	High computational cost and dependency on accurate forecasts
[53]	Bi-objective RL for emergency medical delivery	Balanced cost and speed under emergency conditions	Dual optimization of time and energy	Use-case limited to crisis/emergency logistics
[52]	Q-learning for electric truck–drone routing	Achieved operational cost savings in mixed EV-UAV fleets	Supports sustainable last-mile delivery	Relies on ideal fleet assumptions; lacks real-world validation
[2]	Two-phase UAV path planning with PSO	Reduced drone battery consumption by 15.75%	Improved flight efficiency and resource planning	Assumes static environment; lacks responsiveness
[19]	Multi-objective Q-learning for post-disaster UAV routing	Enhanced equity in relief distribution	Fair and effective supply assignment in humanitarian aid	Ineffective in dynamic multi-agent coordination
[54]	Fuzzy logic with RL for sustainable last-mile delivery	Explored viability of alternative modes (e-bikes, etc.)	Supports eco-friendly logistics mode selection	Restricted by fixed urban infrastructure assumptions

Table 2
Comparative analysis of policy-based MARL approaches for truck–drone logistics.

Study	Methodology	Key contribution	Strengths	Limitations
[47]	PPO-based MARL for truck–drone route optimization	23% route efficiency gain and 17% reduction in operational costs	Enables real-time re-routing and adaptive decision-making	High computational and training demands
[35]	Multi-Agent PPO (MAPPO) for fleet coordination	34% operational efficiency boost, 28% lower delivery time	Scalable to large fleets; robust under urban congestion	Computational load grows exponentially with fleet size
[55]	PPO for real-world drone maneuvering	25% better obstacle avoidance, 19% fewer failed deliveries	Improves navigation in dynamic delivery zones	Requires frequent retraining due to sample inefficiency
[36]	A2C for adaptive truck–drone routing	21% faster delivery, 18% less energy usage	Energy-efficient routing with reliable task allocation	Less effective in highly non-stationary agent environments
[55]	A3C for decentralized fleet decision-making	22% better package distribution, 17% delay reduction	Strong performance in changing logistics networks	Demands high communication bandwidth
[7]	Constrained k-means +deep RL in PLTSPHD model	Higher computational efficiency than heuristic baselines	Reduces planning cost and time	Lacks adaptability for dynamic route updates
[56]	Constraint-Based Hybrid Pointer Network (CH-Ptr-Net) with RL	Improved multi-drone coordination and lower routing cost	Better scalability in multi-agent networks	High memory usage in large-scale settings
[57]	RL for survivable truck–drone delivery under attacks	Enhanced security and operational resilience	Integrates safety with adaptive delivery planning	Needs reliable threat detection models
[5]	Blockchain-integrated policy enforcement RL	Ensures route compliance in urban drone operations	Strengthens trust and data security via smart contracts	Introduces processing overhead from blockchain layers
[58]	Comparative analysis of PPO and DQN for urban logistics	PPO showed better adaptability despite 1.5x training time vs. DQN	More suitable for dynamic environments	High training and inference cost
[32]	PPO-based obstacle-aware UAV path planning	25% flight efficiency gain, 14% battery life extension	Enhances energy-efficient and safe drone navigation	Requires detailed and realistic environmental modeling

5.3.3. Regulatory and safety challenges

Drone-based logistics systems face significant regulatory and safety hurdles, including airspace restrictions, legal compliance, and risk mitigation. To address these, MARL-based compliance-aware models have been introduced, enabling real-time route planning within legal and safety constraints. These frameworks have demonstrated up to 38% improvement in regulatory adherence while maintaining delivery reliability [47,73].

Airspace and Legal Compliance: Wu et al. [10] proposed an adaptive MARL model for real-time airspace monitoring, reducing regulatory violations by 29% and enhancing safe delivery completions by 21%. Garcia et al. [73] implemented a graph-based MARL framework that adjusted drone routes based on real-time legal data, improving compliance by 33% and decreasing route deviations by 19%.

Table 3
Comparative analysis of hybrid learning models in MARL for truck–drone coordination.

Study	Methodology	Key contribution	Strengths	Limitations
[37]	MADDPG-based adaptive truck–drone coordination	35% improvement in delivery time and route flexibility	Enables decentralized execution with centralized training	High computational demand in large-scale scenarios
[31]	Multi-UAV and truck coordination using MADDPG	29% reduction in delivery time, 22% better route reliability	Excels in continuous control and dynamic environments	Limited scalability in urban congestion
[60]	Hybrid RL with time-dependent traffic-aware routing	Enhanced synchronization under urban congestion	Traffic-aware real-time scheduling	Assumes static battery profiles; lacks adaptation to endurance
[61]	MDP-based hybrid RL for scheduled and on-demand tasks	15% profit gain, 50% higher on-demand acceptance	Real-time adaptive coordination and scheduling	High resource requirement for real-time inference
[33]	Deep Q-Learning for UAV-based last-mile delivery	12.8% faster delivery, 8.4% energy gain	Improves route optimization and drone battery use	Less effective in highly dynamic settings
[38]	Hybrid truck–drone with flexible launch/retrieval nodes	Lower operational costs, improved fleet resilience	Flexible deployment and recovery mechanisms	High computational overhead in large-scale use
[62]	Genetic Algorithm +RL for UAV-UGV integration	Faster intervention time and higher decision accuracy	Suited for multi-vehicle operational planning	Poor scalability in real-world logistics
[63]	Multi-visit, multi-launch truck–drone hybrid system	Boosted scalability and reduced costs	Enables delivery from non-customer nodes	Assumes static drone configurations
[40]	Heuristic scheduling with en-route drone operations	30% delivery time savings, 62% drone utilization gain	Improves truck–drone fleet coordination	Lacks adaptability to heterogeneous fleets
[64]	Clustering-based hybrid truck–drone routing for rural areas	Improved coverage and reduced costs in low-density areas	Well-suited to terrain challenges	Difficult to generalize to urban logistics
[65]	Genetic Clustering Algorithm with hybrid RL	Shorter delivery times and improved scheduling	Effective for mid-sized operations	Long processing times for large scenarios
[59]	Meta-RL for generalized truck–drone routing	40% better adaptability, reduced retraining time	Enhances policy transfer and flexibility	Sample inefficiency and high training time
[34]	Meta-RL +Deep Q-learning for adaptive scheduling	32% less computational overhead, 19% better generalization	Improves transferability across logistics settings	Elevated memory and training complexity

Safety Risk Mitigation: Prayitno et al. [74] applied Causal Reinforcement Learning (C-RL) for safe drone path planning, reducing collision risk by 27% and improving public safety adherence by 22%. Gadiraju et al. [31] used federated MARL to facilitate hazard data sharing among drones, enhancing detection accuracy by 31% and reducing safety-related disruptions by 25%.

Cross-Border, Urban, and Disaster Logistics:

- Xu et al. [105] optimized cross-border e-commerce logistics using enhanced Sand Cat Swarm Optimization, though faced challenges in harmonizing international regulations.
- Madani et al. [14] proposed a regulatory framework to bridge industry needs and safety standards, identifying the need for global policy alignment.
- Huang et al. [3] focused on emergency supply delivery, highlighting the necessity of airspace and operational safety under disaster conditions.

Emerging Technologies and Decentralized Safety Models:

- Sabuncu and Bilgehan [106] proposed blockchain-based D2X communication to enhance trust, safety, and regulatory compliance.
- Shao et al. [107] developed an ACO-A* hybrid algorithm for long-range drone routing, emphasizing dynamic compliance with airspace regulations.
- Patchou et al. [24] integrated UAVs into ITSs, highlighting regulatory gaps during the COVID-19 pandemic and proposing system-of-systems coordination.

Safety Integration in Urban and Last-Mile Logistics:

- Zhang et al. [108] proposed a reservation-based two-echelon VRP to optimize urban logistics under legal constraints.
- Shen et al. [109] evaluated zoning vs. no-zoning policies in autonomous truck-robot systems, identifying regulatory compliance as a key cost factor.
- Zhu et al. [110] addressed BEV-Drone routing for rural first-aid delivery, emphasizing battery logistics and compliance in restricted airspace.

Table 4
Comparative analysis of delivery efficiency in MARL-based truck–drone logistics studies.

Study	Methodology	Key contribution	Strengths	Limitations
[47]	MADRL for adaptive truck–drone scheduling	45% delivery time reduction, 30% more successful deliveries	Strong real-time adaptability	High training cost; Simulated
[10]	DRL-based task offloading and routing	38% more efficient deliveries, 22% less idle time	Reduces truck downtime; task-aware planning	Intensive data requirement; Simulated
[31]	UAV-truck coordination via MADDPG	29% faster deliveries, 22% better distribution efficiency	Enhances multi-agent collaboration	Limited to small-scale tests; Simulated
[34]	Q-learning with early arrival path optimization	32% less energy, 95% on-time delivery rate	Balances speed and energy efficiency	Needs tuned reward design; Real-world tested
[69]	Genetic and strawberry plant algorithm optimization	Outperforms CPLEX in large-scale scheduling	Highly efficient metaheuristics	Ignores battery constraints; Simulated
[71]	SOFM for optimal UAV launch site planning	4.4% drone and 1.1% truck distance reduction	Improves launch coordination	Weather-unaware; Real-world tested
[72]	MILP +DRL for congestion-aware routing	Projected $10.49B$ CNY annual savings in Beijing	Suitable for high-congestion settings	Neglects wind/weather factors; Simulated
[8]	Hybrid truck–drone system in regional logistics	Greater efficiency in local deliveries	Realistic travel distance modeling	Poor dynamic reactivity; Simulated
[1]	Metaheuristics for electric and flying vehicles	Better energy use, less travel time	Sustainable logistics focus	Ignores meteorological effects; Simulated
[70]	Dynamic scheduling via Hybrid Genetic Tabu Algorithm	1535% faster deliveries under demand shifts	Works under dynamic demand profiles	Lacks weather/battery modeling; Simulated
[22]	Stochastic Petri Nets for UAV energy modeling	Improved battery usage via recharge-aware paths	Strategic energy modeling	Ignores policy/regulatory context; Simulated
[4]	Multi-objective GA for route optimization	Reduced congestion, improved sustainability	Adapts to evolving objectives	High compute cost for real-time use; Simulated
[38]	Hybrid Q-learning +pruning for underground routing	Improved underground freight delivery	Mitigates surface congestion	Ignores multi-modal integration; Simulated
[41]	Multi-objective RPAS location-routing model	Reduced cost, time, risk, and battery usage	Suitable for emergency logistics	Ignores legal/policy constraints; Simulated
[101]	MINLP-based UAV fleet optimization	Higher revenue, fewer drones used	Optimizes mixed fleet utilization	Lacks environmental consideration; Simulated
[102]	Bayesian Network (BN) for UAV performance metrics	Identifies key delivery impact factors	Strong predictive capabilities	Excludes legal/safety variables; Real-world tested
[73]	G-MARL for adaptive delivery routing	27% route efficiency gain, 18% compute savings	Lowers compute cost for coordination	Needs infrastructure support; Simulated
[42]	Hierarchical RL for clustered fleet coordination	41% better compute efficiency, 35% more throughput	Ideal for large agent networks	Complex model training; Simulated
[74]	Causal RL for uncertain urban conditions	Increases fault tolerance and adaptability	Robust against dynamic traffic	Extra data and training needed; Simulated
[43]	Transfer learning in MARL for adaptability	29% retraining cost reduction, improved regional transferability	Strong cross-environment generalization	Expensive for large domains; Simulated

Scalability and Communication Challenges: As drone fleets expand, real-time legal monitoring becomes computationally intensive. Shahbazian et al. [30] employed Graph Convolutional Cooperative MARL (GCC-MARL) to improve regulatory adaptation by 39% and reduce processing costs by 22%.

Reliable Data Transmission: Regulatory compliance depends on high-speed and secure communication networks. Raja et al. [11] demonstrated that 6G-assisted MARL reduced regulatory update delays by 31% and improved drone coordination by 25%. Lee et al. [43] introduced a situation-aware RL compliance framework, achieving a 37% increase in adherence to urban safety policies.

Future Directions: To advance regulatory-compliant MARL for truck–drone delivery, future work should:

- Develop hybrid MARL architectures combining legal, spatial, and operational constraints.
- Incorporate decentralized airspace monitoring and automated compliance auditing.
- Collaborate with aviation authorities to build standardized airspace models and safety protocols.
- Utilize federated learning and dynamic airspace mapping to ensure safe and scalable multi-agent operations [30,36].

Table 10 compares regulatory and safety in marl-based truck–drone logistics studies.

6. Challenges and future directions

The integration of MARL into truck–drone logistics marks a paradigm shift from conventional optimization techniques, offering dynamic, scalable, and adaptive solutions to complex delivery scenarios. However, the implementation of MARL-based systems remains constrained by several technical, regulatory, and infrastructural challenges that hinder large-scale real-world adoption. These include computational complexity, coordination overhead, regulatory compliance, energy limitations, and integration with heterogeneous environments. To ensure the widespread deployment of MARL-driven systems, future research must advance in hybrid learning frameworks, efficient coordination protocols, and infrastructure-aware optimization strategies.

Table 5
Comparative analysis of energy consumption and battery constraints in MARL-based truck–drone logistics studies.

Study	Methodology	Key contribution	Strengths	Limitations
[47]	MARL-based truck-drone scheduling	35% energy reduction in last-mile delivery	Improved delivery efficiency	Scalability limited by computation complexity
[34]	Q-learning-based early arrival path algorithm	32% energy reduction, 95% on-time delivery	Reliable performance with low energy use	Limited scalability in complex networks
[11]	6G-assisted MARL +autoencoders	28% improved battery lifespan, 21% recharging efficiency	AI-enabled energy-aware planning	Requires 6G infrastructure
[73]	Graph-based MARL (G-MARL)	27% flight range gain, 18% less downtime	Effective energy adaptation	High computation demand
[31]	Federated MARL for UAV-truck coordination	31% energy efficiency improvement	Optimizes recharging logistics	Synchronization issues in dense UAV fleets
[77]	MINLP for charging station optimization	15% logistics cost reduction	Infrastructure cost minimization	Depends on battery tech advancements
[78]	Blockchain fleet sharing for energy-aware routing	Battery life extension and cost savings	Secure, transparent coordination	Relies on multi-party blockchain adoption
[79]	Multi-objective routing (MCRP-DP)	Balanced cost, waiting time, service reliability	Adaptive drone-routing flexibility	Computationally expensive
[76]	Energy models comparing drones, BEVs, and DTs	Drones use 15.8% more energy in wind than BEVs	Realistic cross-mode energy comparison	Limited accuracy under calm conditions
[42]	Graph Convolutional Cooperative MARL (GCC-MARL)	39% faster training, 22% lower compute cost	Scalable MARL deployment	Synchronization bottlenecks
[43]	Situation-aware DRL	26% energy savings, 29% better adaptability	Supports transfer learning	Unproven in live settings
[83]	Truck-and-Drone LRP for CO2 minimization	Lower emissions in last-mile delivery	Sustainable hybrid routing	Urban implementation challenges
[26]	First/last-mile UAV-truck modeling	87% lower energy use vs. trucks	Major environmental gains	Supply chain scale-up limits
[64]	Mobile crowdsensing for UAV coordination	Cost-effective energy-aware UAV logistics	Boosts range and task coverage	Complexity rises with scale
[85]	SA-LKH algorithm for UAV routing	9.67% better quality vs. standard algorithms	Practical for agriculture UAVs	Limited generalizability
[10]	AI-driven battery health monitoring	Prolonged UAV battery lifespan	Promotes operational sustainability	Needs integration with renewables

Table 6
Comparative analysis of computational scalability techniques in MARL-based truck–drone logistics.

Study	Methodology	Key Contribution	Strengths	Limitations
[47]	Decentralized MARL for truck–drone coordination	Achieved 45% improvement in route flexibility	Enables local decision-making; avoids centralized bottlenecks	High computational load with increasing agent count
[42]	Hierarchical MARL architecture	Reduced training overhead by 38%, increased throughput by 33%	Decomposes tasks for better scalability	Difficult to implement in large, heterogeneous fleets
[31]	Hierarchical multi-agent learning	Improved adaptability by 41%, reduced delay by 29%	Efficient agent-role coordination in complex tasks	Complex initial design and role assignment
[73]	Graph-based MARL (G-MARL)	47% improvement in communication, 26% in task allocation	Captures dynamic multi-agent interactions effectively	Expensive graph maintenance in real time
[30]	Graph Convolutional Cooperative MARL (GCC-MARL)	Reduced training time by 39%, computational cost by 22%	Enhanced policy convergence and agent awareness	Needs graph abstraction and tuning expertise
[43]	Meta-Reinforcement Learning (Meta-RL)	34% reduction in training time, 27% increase in generalization	Effective in cross-domain logistics adaptation	Requires large training datasets and pretraining effort
[29]	Mean-Field MARL (MF-RL)	37% stability improvement, 28% reduction in complexity	Scales well for dense agent environments	Less effective in highly non-homogeneous systems
[11]	6G-assisted MARL infrastructure	31% latency reduction, 23% improvement in responsiveness	Ultra-low-latency synchronization for real-time tasks	Requires advanced network availability and deployment
[74]	Causal Reinforcement Learning (C-RL)	Increased agent synchronization by 29%, communication delays cut by 18%	Robust against inter-agent desynchronization	Dependent on high-quality causal modeling frameworks

6.1. Comparative analysis with traditional OR approaches

Traditional OR methods such as exact algorithms (e.g., MILP), heuristics (e.g., Clarke-Wright savings), and metaheuristics (e.g., genetic algorithms, simulated annealing) have been widely used for vehicle routing problems, including truck–drone

Table 7
Comparative analysis of real-world operational coordination in MARL-based truck–drone logistics.

Study	Methodology	Key contribution	Strengths	Limitations
[103]	Blockchain-enabled truck-UAV logistics	Improved routing security, sustainability, and operational cost	Ensures trust and traceability in decentralized systems	Limited scalability due to blockchain latency and data throughput
[86]	Physics-Informed Neural Networks (PINN)	Modeled traffic delays in urban airspace	Integrates physical constraints with learning for better realism	High computational requirements, complex implementation
[89]	YOLOv4 +multi-agent UAV pavement monitoring	Over 95% accuracy in crack/pothole detection	Highly scalable monitoring solution for smart cities	Deployment limited by UAV energy and data relay constraints
[33]	Dynamic replanning using D* algorithm	Improved pathfinding by 92%, reduced collision rate by 5%	Reactive and efficient in dynamic environments	High energy consumption and processor load
[41]	NSGA-II for UAV task scheduling	>30% cost, risk, and battery savings in emergency scenarios	Balances multiple objectives under uncertainty	Complex to scale and adapt to routine logistics
[87]	Vertical delivery drones in tower buildings	Reduced energy consumption and user waiting times	Ideal for confined-space logistics in urban verticals	Not scalable to wide-area or multi-drop logistics
[90]	Data-driven adaptive supply chain management	Proposed sustainable, responsive logistics models	Supports real-time analytics and smart policy adaptation	Alignment complexity with existing infrastructures
[88]	Swarm UAVs with geometric collision avoidance	Enhanced safety in congested urban airspace	Autonomous decentralized behavior modeling	Sensitive to flight altitude and latency instability
[104]	Dual-layer UAV edge +wireless charging	36.46% improvement in task offloading, 39.06% travel distance reduction	Integrates energy provisioning with task coordination	Limited by charger mobility and deployment cost

coordination. These methods offer reliable performance in static environments and are computationally efficient for small to medium problem sizes. However, they typically lack adaptability to dynamic and uncertain real-world conditions such as traffic congestion, weather disruptions, and fluctuating demand.

By contrast, MARL provides decentralized, real-time decision-making capabilities that allow agents (trucks and drones) to learn from environmental feedback and adjust their actions accordingly. Studies have shown that MARL models outperform traditional OR techniques in delivery efficiency, route adaptability, and energy optimization, particularly in highly dynamic logistics settings [10,47,73]. For example, while classical heuristics require predefined schedules and parameters, MARL systems can adapt on-the-fly to new constraints and learn optimal policies over time.

Table 11 offers a dedicated comparison between MARL and traditional OR methods across dimensions such as adaptability, scalability, coordination, real-time responsiveness, and computational requirements. This contrast highlights the practical advantages of MARL in modern logistics networks while acknowledging the computational trade-offs that must still be addressed for large-scale deployment.

6.2. Challenges in MARL-based routing

Despite the advantages, several unresolved challenges continue to hinder MARLs scalability, reliability, and practical deployment. These are summarized as follows:

- **Computational Scalability:** MARL systems face exponential growth in the state–action space as the number of agents increases, resulting in significant computational overhead and slow convergence. Advanced architectures such as GCC-MARL [30] and hierarchical models [42] have mitigated this by 2735%, but real-time scalability remains constrained.
- **Real-Time Coordination:** Low-latency communication and adaptive task allocation are essential for truck–drone collaboration. While 6G-assisted MARL models have improved synchronization by 34% [11], achieving seamless coordination in high-density environments is still difficult due to bandwidth limitations and delayed feedback loops.
- **Regulatory and Safety Compliance:** Varying airspace laws, safety concerns, and lack of harmonized regulations impede MARL deployment. Compliance-aware MARL systems [10,73] have improved adherence by up to 37%, but their generalizability across regions remains limited.
- **Energy Efficiency and Environmental Constraints:** MARL-based drones face significant battery limitations, particularly under adverse weather. Battery-aware MARL models have demonstrated 2431% gains in energy conservation [74], but sustainable routing strategies must address long-term energy usage and emission concerns.
- **Infrastructure and Integration Limitations:** MARL success depends on integration with digital infrastructure such as IoT networks and edge computing. Infrastructural gaps hinder the adoption of situation-aware models, despite showing a 37% boost in delivery reliability [43].
- **Adversarial Robustness and Cybersecurity:** MARL systems are vulnerable to cyber threats including reward hacking and sensor spoofing. Secure learning frameworks have improved robustness by up to 37% [30], but resilient, scalable defenses are required.

Table 8
Comparative analysis of traffic and infrastructure in MARL-based truck–drone logistics studies.

Study	Methodology	Key contribution	Strengths	Limitations
[10]	Attention-based deep RL for dynamic scheduling	39% boost in delivery reliability, 28% drop in congestion delays	Integrates real-time traffic data; improves route flexibility	High computational load; evaluated in simulation only
[73]	Graph-based MARL with traffic-aware re-routing	35% less truck idle time, 22% more successful deliveries	Optimized drone dispatch in urban traffic	Limited scalability in dense city networks; simulated
[74]	Causal RL (C-RL) for infrastructure adaptation	29% better re-routing efficiency, 17% fewer failed deliveries	Airspace compliance under constraints	Dependent on high-resolution infrastructure data
[31]	Federated MARL for distributed traffic response	31% increase in adaptability, 24% reduction in computational load	Agent collaboration with local updates	Requires high synchronization and edge compute capacity
[21]	Cooperative routing with multimodal systems	Bridged theoretical VRP with practical routing in constrained zones	Emphasizes integrated logistics models	Limited endurance and scalability; not field-tested
[60]	Policy-based truck–drone coordination framework	Addressed zoning, airspace, and safety guidelines	Provides regulatory blueprint for urban areas	Depends on regional policy uniformity
[91]	MILP +heuristic for disaster infrastructure routing	Efficient drone-truck collaboration in damaged zones	Real-time damage zone mapping	Computational bottlenecks in real-time events
[92]	Bilevel model with Group Search Optimizer (IGSO)	Improved path planning in AV-UAV logistics networks	Integrates autonomous vehicles with UAVs	Simulation only; lacks live validation
[27]	GIS and maritime-autonomous integration	Reduced fuel and emissions in e-commerce shipping	Geo-spatial routing integration	Real-world scaling and regulatory integration challenges
[18]	VRP with mobile charging via public transport	Cost and energy savings through dynamic recharging	Energy-efficient hybrid infrastructure	Scheduling complexity, lacks field scalability
[67]	Embedded VRP (EMBDD-VRP) for rural delivery	Reduced route complexity and computation time	Suited for 3PL in agriculture logistics	Scalability to urban complexity remains unverified
[93]	DRL +GA for drone route optimization	Lower collisions, better energy efficiency	Reinforces obstacle avoidance	High training overhead and real-time constraints
[68]	Simheuristics and learnheuristics for VRPs	Improved adaptive response to uncertain demand	Real-world traffic simulation capability	Needs dynamic infrastructure feedback integration
[34]	Q-learning with early arrival mechanism	34% computational gain, 21% fewer rerouting failures	Energy-efficient, low-latency decisions	Limited to structured environments; tested in simulation
[11]	6G-enhanced UAV-truck coordination	31% drop in latency, 25% better vehicle synchronization	Ultra-low-latency networking	Infrastructure-heavy; regionally dependent
[43]	Situation-aware RL with mobility-aware routing	37% better delivery rates in changing city traffic	Strong adaptability to urban variability	Pre-training burden; lacks full real-world trials

- **Environmental Sustainability:** Though drones can reduce CO_2 emissions, challenges such as noise pollution and limited recyclability of drone components remain. Wind-aware MARL routing [95] shows promise but needs broader integration into environmental planning models.

Table 12 summarizes the key challenges, proposed MARL-based solutions, and their current limitations, offering insights into the research gaps that must be addressed to scale MARL in real-world truck–drone logistics.

6.3. Future research directions

The future of MARL-based truck–drone logistics hinges on addressing existing challenges while exploring new technological frontiers. Emerging research directions include hybrid learning, cross-domain adaptability, decentralized coordination, edge computing, sustainability, and secure communication. These avenues are essential for developing intelligent, resilient, and scalable logistics systems [36,47].

- **Hybrid Learning Models for Scalable Efficiency:** MARL systems face computational bottlenecks in real-time environments. Hybrid learning-combining heuristics, rule-based reasoning, and DRL-has shown promise in reducing decision latency by 31% while maintaining high success rates [73]. Meta-learning further enhances this by improving convergence speed by 26% and reducing energy consumption by 19% [74]. Future work should focus on lightweight hybrid models optimized for real-world constraints.

Table 9
Comparative evaluation of environmental and weather-adapted approaches in MARL-based truck–drone logistics.

Study	Approach/Algorithm	Key contributions	Strengths	Limitations
[10]	Attention-based Deep RL (ADRL)	+39% flight stability, −28% power loss in harsh weather	Dynamic re-routing using real-time meteorological inputs	High training complexity; limited scalability
[73]	Graph-based MARL (G-MARL)	−35% emergency landings, +22% battery efficiency	Robust against wind fluctuations, optimized re-routing	Computationally expensive
[74]	Causal RL (C-RL)	+31% energy savings, +27% extended flight range in storms	Real-time, battery-aware policy adaptation	Sensor data dependency
[31]	Federated MARL	−24% drone power failures via collaborative flight planning	Distributed learning; improved drone coordination	Needs stable communication networks
[95]	Wind-aware FSTSP extension	CO_2 and energy consumption reduction	Sustainability focus; wind-aware path planning	Synchronization and noise pollution issues
[17]	IoT-assisted logistics management	+97.8% operational efficiency; +98.9% user satisfaction	High performance via sensor-based reactivity	Limited scalability across heterogeneous systems
[98]	UAV-fleet integration with policy roadmap	Strategic adoption framework for UAV use in mixed fleets	Disaster relief support; structured deployment strategy	Safety and regulatory limitations
[96]	Multi-depot VRP with UAVs	Effective coordination for humanitarian response logistics	Fast dynamic reallocation under disaster demands	High computational complexity in large deployments
[16]	Artificial Bee Colony (ABC) algorithm	Reduced fleet size and deployment time in emergencies	Energy-efficient disaster logistics	Weakness under high uncertainty levels
[56]	A-Ptr-Net trajectory optimization	Adaptive drone path planning in weather-variable scenarios	High spatial accuracy; efficient route generation	Energy sensitivity and computational load
[99]	Bayesian Coalition Framework (BCoSDD)	Improved cost-efficiency in stochastic scheduling	Resilient planning under uncertainty	Weakness in real-time adaptation
[97]	UAV-based emergency medical support	Faster medical response in critical scenarios	Supports time-sensitive delivery for public health	Airspace restrictions limit practical reach
[94]	ML-based energy modeling	Enhanced UAV energy prediction accuracy	Enables proactive flight scheduling and recharging	Underperforms in wind-volatile settings
[28]	Hybrid truck–drone control using LQR	Flexible delivery control in changing conditions	Adaptive decision-making and real-time responsiveness	Complex to coordinate synchronously
[75]	Analytical energy efficiency framework	Benchmark trade-off analysis for drones and trucks	Supports sustainable logistics design	High dependency on accurate demand estimation
[34]	Q-learning for wind-adaptive routing	+34% computational efficiency; −21% delivery delay	Lightweight and energy-aware learning	Not yet validated on physical hardware
[11]	6G-enhanced UAV-truck MARL model	−31% latency, +25% coordination under changing weather	Ultra-low-latency coordination with edge learning	Costly infrastructure dependency
[43]	Situation-Aware RL (SA-RL)	+37% delivery success under extreme conditions	Adaptability to dynamic weather input	Requires extensive training datasets

- **Transfer Learning for Cross-Environment Adaptation:** Traditional MARL requires retraining in new environments, limiting flexibility. Transfer learning enables policy reuse, improving model adaptability by 27% and reducing training costs by 38% [43]. Federated transfer learning further reduces adaptation overhead by 34% [10]. Future research should explore domain-agnostic models capable of adapting across urban, rural, and emergency scenarios.
- **Blockchain-Enabled Secure Communication:** Secure, tamper-proof agent coordination is vital in distributed MARL systems. Blockchain integration enhances cybersecurity by 41% and reduces fraud risk by 28% [11]. Smart contract-based trust verification has also improved route security by 32% [34]. Future research should explore lightweight blockchain-MARL integration to support secure decentralized operations.
- **Human-AI Collaboration and Interpretability:** Combining human expertise with AI automation ensures reliability in complex logistics. Human-in-the-loop MARL has improved routing efficiency by 36% and reduced failure rates by 22% [31]. Explainable AI (XAI) has boosted stakeholder trust by 31% [30]. Future work should focus on intuitive interfaces and real-time explainability frameworks, especially for high-risk scenarios.
- **Decentralized and Federated MARL Architectures:** Centralized training is a bottleneck in large-scale deployments. Decentralized MARL and federated learning reduce communication overhead by 29% while preserving coordination accuracy [36]. Hierarchical MARL has improved multi-agent efficiency by 35% [42]. Scalable decentralized architectures should be a priority for future fleet-level operations.

Table 10
Comparative analysis of regulatory and safety in MARL-based truck–drone logistics studies.

Study	Methodology	Key contribution	Strengths	Limitations
[10]	Adaptive MARL for real-time airspace monitoring	29% reduction in violations, 21% increase in safe delivery rates	Improves dynamic airspace compliance	High computational demands, limited scalability
[73]	Graph-based MARL for legal compliance	33% boost in regulatory adherence, 19% fewer route deviations	Adaptive route regulation, scalable in mid-scale networks	High resource requirements in urban deployment
[74]	Causal RL for safe routing under regulations	27% fewer collisions, 22% improvement in safety adherence	Enhances public safety, integrates legal rules dynamically	Relies on real-time infrastructure and data accuracy
[31]	Federated MARL for shared airspace control	31% better hazard detection, 25% fewer safety disruptions	Promotes multi-agent collaboration, data decentralization	Requires robust infrastructure, challenging in rural areas
[105]	Sand Cat Swarm Optimization with enhanced weights	Cost, emission, and satisfaction improvements in cross-border e-commerce	Balances regulatory, environmental, and customer priorities	Complex to generalize across regions with distinct policies
[14]	Policy framework for drone regulatory integration	Structured recommendations for legal and safety alignment	Bridges industrial practice and policymaking	Regional inconsistencies, needs stakeholder collaboration
[24]	System-of-systems model for UAV logistics	Increased capacity and reduced human intervention during pandemics	Prioritizes critical deliveries, improves safety	Inefficient for non-critical deliveries, regulatory fragmentation
[3]	OPTICS-FCM with adaptive genetic algorithm	Reduced delivery time/costs in emergency supply chains	Effective in disaster logistics, spatial optimization	Complex coordination and deployment in real time
[106]	Blockchain +5G-based secure drone communication	Improved D2X safety and fraud prevention	Secure, real-time compliant communication	Costly to scale, regulatory adoption barriers
[107]	Ant colony +A* for long-distance drone compliance routing	Optimized multi-hop pathfinding with minimal violations	Energy-efficient, safe long-range UAV delivery	Challenges in adapting to dynamic airspace constraints
[108]	Two-echelon vehicle routing +bay reservation	Lower emissions and congestion, improved compliance coordination	Enhances intermodal logistics and scheduling	Complexity in urban multi-operator settings
[109]	Autonomous mobile robot system with zoning policies	Cost-effective delivery throughput optimization	Flexible for last-mile logistics zones	Synchronization and real-time regulatory updates required
[110]	BEV-TSP with drone for rural first-aid delivery	Fast emergency delivery, optimized energy use	Improved coverage in remote areas	Scalability, rural airspace compliance, energy logistics
[30]	GCC-MARL for regulatory adaptation	39% faster adaptation, 22% lower computational overhead	Real-time compliance monitoring, scalable coordination	Graph embedding complexity, large-scale network deployment
[11]	6G-enabled UAV-truck MARL architecture	31% less delay in updates, 25% better coordination accuracy	Fast, secure communication for compliance tasks	Dependent on emerging 6G infrastructure, costly setup
[43]	Situation-aware MARL with compliance framework	37% rise in dynamic adherence to safety protocols	Adapts to urban policy variability, improves response	Requires high training data, difficult in low-data regions

- **Edge Computing for Real-Time Adaptability:** Cloud-based inference introduces latency. Edge computing brings decision-making closer to the source, improving response times by 28% and decision accuracy by 22% [34]. Research should explore MARL-edge integration to support ultra-low latency in high-traffic environments.
- **Energy-Aware and Sustainable MARL Frameworks:** Energy constraints limit UAV endurance. Green MARL models have improved energy efficiency by 37% and extended flight time by 31% [74]. Future directions include integrating renewable energy sources, battery health diagnostics, and eco-routing into MARL models for sustainable logistics.
- **Adversarial Robustness and Cybersecurity:** As drones become networked assets, MARL systems are vulnerable to cyber threats. Secure learning frameworks have reduced attack vulnerabilities by 37% and improved resilience by 26% [42]. Future work should develop adversarial training, robust policy shielding, and intrusion detection mechanisms.
- **Smart City Integration and Interoperability:** MARL must interface with urban IoT, 6G networks, and real-time traffic systems. Situation-aware MARL models have improved delivery reliability by 37% through better infrastructure alignment [43]. Future research should develop interoperable standards for MARL integration into smart mobility ecosystems.
- **XAI for Transparent Logistics:** As MARL decisions grow in complexity, transparency becomes critical. XAI can demystify policy behaviors, especially in high-stakes applications like emergency response. Efforts should focus on real-time visualizations, causal explanation models, and stakeholder-specific interpretability layers.

Table 11
Comparison of MARL and traditional OR methods for truck–drone routing.

Criteria	Traditional OR methods (e.g., MILP, GA, SA)	MARL-based approaches
Adaptability to Dynamic Environments	Limited adaptability requires re-optimization with changing conditions	High adaptability learns and updates policies based on real-time feedback
Scalability with Agent Count	Computationally efficient for small- to mid-scale problems but suffers under large-scale scenarios	Scalable through decentralized frameworks (e.g., federated MARL, graph-based MARL)
Real-Time Decision-Making	Typically slow relies on precomputed solutions and heuristics	Capable of real-time adaptation using learned policies and state feedback
Multi-Agent Coordination	Requires centralized scheduling or rule-based decomposition	Enables decentralized coordination through reward sharing and policy alignment
Computational Overhead	Lower during execution but higher for dynamic re-optimization	Higher during training but efficient at runtime post-convergence
Robustness to Uncertainty	Less robust struggles with noisy, incomplete, or uncertain data	Robust learns to handle stochastic and partially observable environments
Explainability and Control	High decision paths are traceable via rules and optimization models	Lower requires XAI add-ons to interpret policies
Integration with IoT/Smart Systems	Manual data integration required	Seamlessly integrates with IoT, sensors, and edge computing platforms
Energy and Sustainability Awareness	Limited incorporation of energy models unless explicitly included	Enables energy-aware routing via reward shaping and policy training

- **Multi-Objective Optimization for Logistics Trade-offs:** Logistics involves competing objectives-minimizing time, cost, emissions, and risk. Multi-objective MARL can balance these dynamically. Future research should formalize reward shaping and Pareto-efficient decision-making in MARL to optimize trade-offs across heterogeneous environments.

In summary, advancing MARL-based truck–drone logistics requires interdisciplinary innovation across learning architectures, communication security, human-AI interaction, and sustainability. By integrating XAI, edge intelligence, federated coordination, and transfer learning, future systems can achieve scalable, secure, and eco-efficient operations [36].

7. Conclusion

MARL has emerged as a pivotal framework for enhancing truck–drone logistics, offering solutions to complex challenges such as dynamic task allocation, real-time routing, energy optimization, and regulatory compliance. This survey has demonstrated how MARL, through approaches like DQN, MAQL, PPO, A2C, and MADDPG, enables intelligent coordination and adaptability in highly dynamic delivery environments. Empirical studies consistently show that MARL models outperform traditional heuristics in delivery efficiency, energy conservation, and route adaptability-highlighting MARL's potential to transform last-mile logistics networks.

Despite its promise, several challenges continue to impede MARL's full-scale deployment. High computational overhead, communication latency in decentralized networks, regulatory variability, and limited energy resources pose barriers to scalability and real-time responsiveness. Moreover, issues such as adversarial robustness, integration with smart city infrastructure, and explainability of agent decisions remain underexplored. To address these limitations, recent advancements have introduced hybrid learning architectures, graph-based policy networks, 6G-enabled communication frameworks, and blockchain-driven coordination systems. These developments mark a significant step toward operational scalability and system resilience in real-world applications.

Looking forward, the future of MARL in truck–drone logistics lies in advancing lightweight, decentralized, and human-interpretable models. Key research priorities include hybrid MARL architectures that integrate DRL with heuristic search for computational efficiency, federated learning for decentralized policy updates, and edge computing for ultra-low-latency decision-making. Incorporating XAI into MARL frameworks will be vital for improving human trust, regulatory transparency, and ethical accountability. Furthermore, sustainability-focused MARL models that optimize energy use and reduce carbon emissions will be central to building eco-friendly logistics systems in smart cities and remote regions alike.

In summary, MARL-based truck–drone logistics represents a transformative paradigm for next-generation delivery networks. By addressing the current challenges through interdisciplinary innovation and technological integration, MARL has the potential to enable intelligent, adaptive, and sustainable logistics solutions. Continued research into robust, secure, and scalable MARL frameworks will be crucial for realizing fully autonomous and trusted logistics infrastructures in diverse real-world contexts.

Declaration of competing interest

Authors declare no conflict of interest.

Acknowledgment

The author, Ali Arishi, at King Khalid University would like to acknowledge the support by the Deanship of Scientific Research through King Khalid University, Saudi Arabia funded by the Large Group Research Project RGP2/267/46.

Table 12
Summary of key challenges and limitations in marl-based truck–drone logistics.

Challenge	Description	Proposed solutions	Limitations	Impact
Computational Overhead	MARL models face exponential state–action space growth, leading to high training costs (up to 40% more than centralized RL) [10].	Graph-based MARL (GCC-MARL) reduces complexity by 32%, hierarchical MARL lowers training overhead by 35% [30,42].	High-performance hardware is still required; real-time adaptation is difficult at scale.	Delays deployment in large-scale logistics due to long training cycles and high resource demand.
Communication Constraints	Continuous state information sharing is hindered by network latency and bandwidth issues [31].	6G-assisted MARL reduces synchronization delays by 34%, enhances coordination accuracy by 28% [11].	Dependence on emerging 6G infrastructure; real-time stability in dense areas is uncertain.	Poor communication leads to suboptimal coordination and increased delivery times.
Regulatory Compliance	Fragmented drone airspace regulations limit routing flexibility across regions [74].	Real-time regulatory-aware MARL improves compliance by 37%, graph-based models enhance legality by 33% [10,73].	Compliance models lack generalizability; no global regulatory standard exists.	Legal inconsistencies disrupt operations and delay large-scale implementation.
Safety and Collision Avoidance	UAVs must avoid obstacles in dynamic, congested environments [31].	Federated MARL improves hazard detection by 31%, reduces collision risks by 25% [74].	Safe path planning across fleets remains computationally intensive.	Accidents compromise public safety, increase liability, and reduce trust in drone systems.
Real-Time Decision-Making	High-frequency environmental changes require rapid rerouting and task allocation [36].	Situation-aware RL improves adaptability by 41%, reduces decision delays by 29% [43].	Real-time responses still lag under large-scale operational stress.	Delays degrade service quality and increase operational costs.
Cybersecurity and Robustness	Sensor spoofing, policy tampering, and communication disruption threaten system integrity [30].	Secure MARL reduces attack vulnerabilities by 37%, boosts system resilience by 26% [42].	Requires continuous updates and context-aware security layers.	Threats may halt services, compromise data, and damage system reliability.
Energy Constraints	UAVs face limited battery life, especially under adverse conditions [47].	Battery-aware MARL increases energy efficiency by 31%, extends flight range by 27% [74].	UAV endurance and recharging infrastructure still limit long-haul delivery.	Energy limitations lead to delivery failures, downtime, and logistical inefficiencies.
Scalability	Adding agents increases coordination complexity exponentially [30].	Q-learning path optimization improves computational efficiency by 34% [34].	Performance deteriorates in large, congested networks.	Inhibits expansion and limits real-world deployment at city scale.
Smart Infrastructure Integration	Inconsistent IoT infrastructure hinders synchronization with smart city systems [36].	SA-RL improves delivery success by 37% via contextual awareness [43].	Interoperability issues with existing city systems persist.	Reduces routing efficiency, affects adaptability in heterogeneous environments.
Environmental Sustainability	Drone operations contribute to noise, energy use, and emissions [95].	Wind-aware and energy-optimized routing reduce emissions and operational costs.	Noise impact and sustainability metrics under dynamic conditions remain unresolved.	Environmental concerns limit regulatory approvals and long-term public acceptance.

Data availability

No data was used for the research described in the article.

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